

Chamodi Karunaratne

Software Engineering Intern

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ABOUT ME

Information Technology and Management undergraduate at the University of Moratuwa, focused on full-stack engineering and systems design using Java, Next.js, React. Passionate about blending robust backend logic with creative, high-end UI/UX prototyping. Balancing strong academic principles with hands-on experience in requirements engineering and IoT frameworks to build highly optimized, out-of-the-box digital solutions.

EDUCATION

BSc. (Hons) Information Technology & Management 2024 – Present

Faculty of IT, University of Moratuwa 

- **CGPA:** 3.78 / 4.00
- **Academic Honors:** Included in the Dean's List for Semester 01

G.C.E. Advanced Level, Devi Balika Vidyalyaya, Colombo 08 2020 – 2023

- 2022(2023) - Commerce Stream (English Medium)

TECHNICAL SKILLS

Languages: Java, Typescript, JavaScript, C, C++, SQL, Python

Frameworks & Libraries: Tailwind CSS, Next.js, React, Node.js, Spring Boot, REST APIs

Databases & ORMs: MySQL, PostgreSQL, Drizzle, Hibernate, Spring Data JPA, Firebase Realtime DB, Neon

DevOps Tools: Git, Github, Docker, Vercel, Expo

Hardware Platforms: ESP32, Arduino, PCB Design (EasyEDA)

PROJECTS

Enterprise IT Asset Management System (EITAMS)  11/2025 – 09/2026

Full-Stack Software Developer

- Built a scalable, enterprise-grade Next.js/React asset registry featuring complex data grids, real-time synchronization, and automated PDF tag generation.
- Designed relational databases with Drizzle ORM (PostgreSQL/MySQL) to securely manage bulk asset transfers, reporting, and NextAuth role-based access.
- Improved codebase maintainability by refactoring monolithic UI files into modular React hooks while collaborating in an Agile team environment.
- **Tech Stack:** Next.js, React, TypeScript, Tailwind CSS, PostgreSQL, Drizzle ORM, Git, GitHub

Personal Styling & Outfit Recommendation Application  09/2026 – Present

Full-Stack Developer

- Developing a context-aware personal styling platform that recommends outfits, shoes, accessories, makeup, and hairstyles based on occasion, weather, location, transport, and user preferences.
- Building a responsive Next.js frontend and Java Spring Boot backend with a modular architecture for user profiles, occasion planning, personalized recommendations, and saved looks.
- Designing rule-based recommendation logic with planned LLM-assisted features and external integrations such as weather services to generate practical, personalized styling suggestions.
- **Tech Stack:** Next.js, React, TypeScript, Tailwind CSS, Java, Spring Boot, Spring Data JPA, Hibernate, REST APIs, Git, GitHub

Skyforge - Environmental Monitoring Drone & Real-Time Dashboard

12/2024 – 08/2025

PCB Designer & Frontend Developer

- Designed a custom EasyEDA PCB for an ESP32-based environmental monitoring drone, integrating sensors and communication modules.
- Developed a responsive React dashboard with Firebase Realtime Database integration for real-time environmental and flight telemetry monitoring.
- Built interactive data visualizations for weather parameters, GPS, altitude, air quality, and flight status.
- **Tech Stack:** React, TypeScript, Tailwind CSS, Firebase Realtime Database, Recharts, ESP32, EasyEDA, Arduino IDE, C++, Git, GitHub

Woof - Pet Adoption System

10/2023 – 04/2024

- Developed a Java-based desktop pet adoption management system using Swing and AWT event handling.
- Implemented role-based authentication, MySQL integration, and CRUD operations including report generation using JDBC.
- Built user-friendly interfaces with search, reporting, file handling, PDF generation, and i18n support.
- **Tech Stack:** Java, Swing, AWT, MySQL, JDBC, i18n, File I/O, PDF Generation, Git, GitHub

CERTIFICATES

BCS Level 4 Certificate in IT

Java Programming Certification
SLIIT

EXPERIENCE

IoT & Embedded Systems Labs (IES Labs), Software Developer

09/2025 – Present

• Electronic Team Member

ICTBUS Institute, Academic Instructor

01/2025 – 01/2026

COMMUNITY AND LEADERSHIP

Hackelite 2.0, IEEE WIE Student Branch Affinity Group of UOM

Lead - Finance Committee

Annual General Meeting 2025, IEEE Professional Communication Society

Lead - Logistics Handling

Road to Legacy 2.0, IIEE of USJ

Member - Program Committee

Binara Padura 2.0, Rotaract Club of University of Moratuwa

Member - Finance Committee